

Math 215B. Spring 2024. Homework answers

1. Which of the following spaces are pairwise homeomorphic and which are not: (a) the orthogonal group SO_3 , (b) the space $T_1 S^2$ of unit tangent vectors to S^2 , (c) the Stiefel manifold $V(3, 2)$, (d) in the complex 3-space $\mathbb{C}^3$ with coordinates z_1, z_2, z_3 , the set of unit vectors (i.e. $|z_1|^2 + |z_2|^2 + |z_3|^2 = 1$) satisfying $z_1^2 + z_2^2 + z_3^2 = 0$, (e) the real projective space $\mathbb{R}P^3$, (f) $S^2 \times S^1$?

Answer: All except (f) are homeomorphic to each other. In (a,b,c), each space consists of ordered pairs of perpendicular vector of fixed lengths (in the SO_3 case, a matrix's third column is the cross-product the the first two). In (d), writing each z_i as $x_i + \sqrt{-1}y_i$ we obtain the relations $\sum x_i^2 + \sum y_i^2 = 1$, $\sum x_i^2 - \sum y_i^2 = 0$, and $2 \sum x_i y_i = 0$, which lead to the same description: a pair $(\vec{x}, \vec{y})$ of orthogonal vectors of length $1/\sqrt{2}$ each.

The identification of $\mathbb{R}P^3$ with SO_3 is famous but less straightforward. Consider the action of the group $Sp_1 = S^3$ of unit quaternions on $\mathbb{H}$ by conjugations: $x \mapsto uxu^{-1}$. It leaves the 3-dimensional space of $bi+cj+dk$ of purely imaginary quaternions invariant and acts on it by (length-preserving and hence) orthogonal transformations, which defines a homomorphism $Sp_1 \rightarrow SO_3$. It is not hard to see that it is surjective, with the kernel $\pm I$. Thus, SO_3 is identified with $S^3/(\pm 1) = \mathbb{R}P^3$.

It is *a priori* unclear how to prove that two spaces are *not* homeomorphic. But we know the fundamental groups $\pi_1 S^2 \times S^1 = \mathbb{Z}$ and $\pi_1 \mathbb{R}P^3 = \mathbb{Z}_2$.

2. In the subsection 5.4D (pages 48-49) the partition of flag manifolds into cells (known as Bruhat cells) is defined. (a) For the manifold of complete flags $V^1 \subset V^2 \subset \mathbb{C}^3$ in (say, complex) 3-space (actually the field doesn't matter), describe explicitly which flags belong to which cell (the total number should be 3!) (b) Let c_k be the number of k -dimensional (in complex units) Bruhat cells in the manifold of complete flags in an n -dimensional complex space. Show that the generating function $c_0 + c_1 q + c_2 q^2 + \dots$ of this sequence is equal to the " q -factorial": $\prod_{k=1}^n (1 - q^k)/(1 - q)$, and check this for your answer in (a). Hint: Consider the Bruhat cell partition of the manifold of complete flags in the space F^n , where F is a finite field with q elements.

Answer. (a) Let $\mathbb{C}^1 \subset \mathbb{C}^2 \subset \mathbb{C}^3$ be the reference flag. The 0-dimensional cell is the flag with $V^1 = \mathbb{C}^1, V^2 = \mathbb{C}^2$. There are two 1-dimensional cells: one with $V_2 = \mathbb{C}^2$ and V^1 any line in $\mathbb{C}^2$ other than $\mathbb{C}^1$, the other with $V_1 = \mathbb{C}^1$ and V^2 any plane containing $\mathbb{C}^1$ other than $\mathbb{C}^2$. There are two 2-dimensional cells: one with $V^2 \neq \mathbb{C}^2$ containing $\mathbb{C}^1$ and V_1 any line in V^2 other than $\mathbb{C}^1$, the other with V_1 any line in $\mathbb{C}^2$ other than $\mathbb{C}^1$ and V_2 any plane containing

that line other than $\mathbb{C}^2$. There is one 3-dimensional cell: with V_1 any line not contained in $\mathbb{C}^2$ and V_2 any plane containing V_1 and not containing $\mathbb{C}^1$.

(b) A complete flag in F^n is specified by one of $q^n - 1$ non-zero vectors $v \in F^n - 0$ modulo $q - 1$ rescalings by non-zero scalars, and by a complete flag in the quotient space $F^n / \text{span}(v)$. We conclude that the number f_n of complete flags in F^n is equal to $f_{n-1}(q^n - 1)/(q - 1)$, and by induction $f_n = \prod_{k=1}^n (q^k - 1)/(q - 1)$. On the other hand, each Bruhat cell of dimension k is parameterized by points of an affine k -dimensional space F^k which consists of q^k elements. If c_k is the number of such cells, then $f_n = \sum c_k q^k$ as required. In particular $f_3 = (q^3 - 1)(q^2 - 1)(q - 1)/(q - 1)^3 = (1 + q + q^2)(1 + q) = 1 + 2q + 2q^2 + q^3$ in agreement with part (a).

3. Prove that a CW-complex is contractible if it is the union of an increasing sequence $X_1 \subset X_2 \subset \dots$ of subcomplexes such that each X_n is contractible in X_{n+1} (and conclude that, in particular, S^∞ is contractible).

Answer. On the time interval from 0 to $1/2$, contract X_1 inside X_2 to a point $x_0 \in X_1$ and extend this homotopy to the whole of X using Borsuk's theorem. Then on the time interval from $1/2$ to $3/4$, contract X_2 inside X_3 to x_0 and extend this homotopy to the whole of X^1 , and so on. The composition of these homotopies yields a map $X \times [0, 1) \rightarrow X$ which extends continuously to $X \times [0, 1] \rightarrow X$ (by the W axiom), which on $X \times 1$ is the constant map $X \rightarrow x_0$ (thus providing a deformation retraction of X to x_0). The argument applies to S^∞ since S^n is contractible inside S^{n+1} .

P.S.: Correction. At it was noted by one student, the previous argument would be correct if the point x_0 stays fixed during each stage of the homotopy (which in the example of S^∞ is easy to arrange), but otherwise has no reason to extend continuously to the final time moment $t = 1$ of the homotopy.

Here is one way to fix this. Introduce space $\tilde{X}$ obtained by attaching to X the interval $[0, 1]$ by identifying 1 with x_0 , and declare 0 the base point of $\tilde{X}$ (which is homotopy equivalent to X since the interval can be deformationally retracted to x_0). Whenever we are given a contraction of $X_n \times I \rightarrow X_{n+1}$ to a point (say, $x_1 \in X_{n+1}$), extend this homotopy to $\tilde{X}_n$ by gradually stretching the interval $[0, 1]$ along the trajectory of x_0 on its way to x_1 , so that at the last moment $[0, 1/2]$ is stretched to $[0, 1]$ and $[1/2, 1]$ parameterizes the trajectory from x_0 to x_1 . After that, continue the homotopy by shrinking the path from

¹i.e. compose this extended homotopy $X \times [1/2, 3/4] \rightarrow X$ with $f_{1/2} \times \text{id}_{[1/2, 3/4]}$, where $f_{1/2} : X \rightarrow X$ is the map resulted from the previous homotopy and sending $X_1 \rightarrow x_0$

0 through $1 = x_0$ to x_1 back to the base point 0. This results in a contraction of $\tilde{X}_n$ inside $\tilde{X}_{n+1}$ to the base point 0 which keeps the base point fixed all the time.

4. Prove that the image of the canonical homomorphism $\pi_2(X, x_0) \rightarrow \pi_2(X, A, x_0)$ lies in the center of $\pi_2(X, A, x_0)$.

Answer. Figure 38 on page 94 shows how to interchange the order in the sum $f + g$ of two absolute spheroids. The same homotopy works when the top side of the rectangle labeled by g is mapped to A (rather than the base point x_0) provided that the top side of the rectangle labeled by f is still mapped to x_0 .

5. Let X be the space obtained by attaching one end of a 2-dimensional cylinder to the other end by the double cover map of the circle. (This space is sometimes called the "mapping torus" of the double-cover map of S^1 to S^1 .) Compute all homotopy groups of X .

Answer. Describing the space as a bouquet of two circles a, b with a 2-cell attached, we find the presentation of the fundamental group π_1 by the relation $aba^{-1}b^{-2} = e$. Note that $b = (a^{-1}ba)^2 = (a^{-2}ba^2)^4 = \dots$, i.e. π_1 contains a normal subgroup isomorphic (in its additive notation) to binary rationals $\mathbb{Z}[1/2] = \{m/2^k\}$. The quotient by this subgroup (described in generators and relations by declaring $b = e$) is the infinite cyclic group generated by a . Thus, π is the semidirect product of $\mathbb{Z}$ with $\mathbb{Z}[1/2]$ corresponding to the action of a (resp. a^{-1} by conjugations on $\mathbb{Z}[1/2]$ as the multiplication (resp. division) by 2.

The regular $\mathbb{Z}$ -covering of X corresponding to this normal subgroup is obtained by gluing together a two-sided infinite sequence of cylinders $S^1 \times I$ by attaching the right end of a previous cylinder to the left end of the next one via the double-cover map $S^1 \rightarrow S^1$. A spheroid (of dimension > 1) mapped to this covering space lands in a finite collection of such cylinders, which can be deformationally retracted to its rightmost end S^1 where the spheroid is contractible (since the universal cover of S^1 is). This, $\pi_n(X) = 0$ for $n > 1$.

6. Prove that fibers of a Hurewicz fibration over a path-connected base are homotopy equivalent to each other.

Answer. One way would be to repeat the proof of WHE between the fibers F_x of a Serre fibration $E \xrightarrow{p} B$ with a path-connected base replacing maps $Z \rightarrow F_x$ of CW-complexes Z with maps arbitrary Z .

Alternatively, and more directly, consider the identity map $F_{x_0} \rightarrow F_{x_0}$ and a homotopy $F_{x_0} \times [0, 1] \xrightarrow{pr_2} [0, 1] \xrightarrow{\gamma} B$ defined by a path $\gamma : [0, 1] \rightarrow B$ connecting x_0 with x_1 . By the Covering Homotopy Property, there is a map $F_{x_0} \times [0, 1] \rightarrow E$ covering this path and thus defining a map $f : F_{x_0} \rightarrow F_{x_1}$ covering end of the homotopy. Likewise, the reverse path γ^{-1} gives rise to a map $g : F_{x_1} \rightarrow F_{x_0}$, while the composite paths $\gamma\gamma^{-1}$ and $\gamma^{-1}\gamma$ result in the compositions $g \circ f$ and $f \circ g$ respectively, each obtained from composing the two covering homotopies and taking the last map of the composition. On the other hand, each composite path is a loop (based at x_0 and x_1 respectively) contractible to the constant loop at the base point. Applying the Covering Homotopy Property to such contractions we obtain homotopies $g \circ f \sim \text{id}_{F_{x_0}}$ and $f \circ g \sim \text{id}_{F_{x_1}}$.

For example, one of the composed homotopies gives us a map $F_{x_0} \times [0, 2] \rightarrow E$ covering the loop $t \mapsto \gamma(t)$ for $0 \leq t \leq 1$ and $t \mapsto \gamma^{-1}(t - 1)$ for $1 \leq t \leq 2$, which at $t = 0$ and $t = 2$ is $\text{id}_{F_{x_0}}$ and $g \circ f : F_{x_0} \rightarrow F_{x_0}$ respectively. Contracting this loop we obtain a homotopy map $[0, 2] \times [0, 1] \rightarrow B$ which sends to x_0 the union I of three sides of the square: $0 \times [0, 1]$, $[0, 2] \times 1$, and $2 \times [1, 0]$. Applying the Covering Homotopy Property to this homotopy we obtain a covering map $F_{x_0} \times [0, 2] \times [0, 1] \rightarrow E$ whose restriction to I provides a homotopy $F_{x_0} \times I \rightarrow F_{x_0}$ between $\text{id}_{F_{x_0}}$ and $g \circ f$.

7. Compute $\pi_2(S^1 \wedge S^2)$ and the action of $\pi_1(S^1 \wedge S^2)$ on it.

Answer. Obviously the fundamental group of the space is $\mathbb{Z} = \pi_1(S^1)$ which comes from the embedding of S^1 into it. The universal $\mathbb{Z}$ -cover can be described as the real line $\mathbb{R}$ with a sequence of spheres S^2 attached at the integer points of $\mathbb{R}$ and the deck transformation group acting by integer translations along $\mathbb{R}$. Since $\mathbb{R}$ is contractible, the space is homotopy equivalent to the quotient by it, i.e. to the bouquet $\bigwedge_{n \in \mathbb{Z}} S^2_{(n)}$ of countably many 2-spheres. Therefore $\pi_2 = \bigoplus_{n \in \mathbb{Z}} \mathbb{Z}$, the groups of 2-sided sequences $(\dots, x_{-1}, x_0, x_1, x_2, \dots)$ of integers with finitely many non-zero entries each, and the action of π_1 is given by translations of the sequences: $a \in \mathbb{Z} = \pi_1$ maps $\{x_n\}$ into $\{x_{n+a}\}$.

8. Compute the 3rd homotopy group of unitary groups U_n for all n . *Hint:* Use the fibration of U_2 over U_1 defined by the determinant function, and the fibrations of U_n over S^{2n-1} defined by projection of a unitary matrix to its 1st column.

Answer: $\pi_3(U_n) = \mathbb{Z}$ for $n > 1$ and $\pi_3(U_1) = 0$. Indeed, from the long

exact sequence of fibration $\det : U_2 \rightarrow U_1$ with the fiber $SU_2 \cong S^3$, we have

$$0 = \pi_4(U_1) \rightarrow \pi_3(S^3) \rightarrow \pi_3(U_2) \rightarrow \pi_3(U_1) = 0,$$

and hence $\pi_3(U_2) = \pi_3(S^3) = \mathbb{Z}$. From the long exact sequence of the fibration $U_{n+1} \xrightarrow{U_n} S^{2n+1}$, we have

$$\pi_{q+1}(S^{2n+1}) \rightarrow \pi_q(U_n) \rightarrow \pi_q(U_{n+1}) \rightarrow \pi_q(S^{2n+1}),$$

and hence $\pi_q(U_n) = \pi_q(U_{n+1})$ for $q < 2n$. In particular,

$$\mathbb{Z} = \pi_3(U_2) = \pi_3(U_4) = \pi_3(U_5) = \cdots$$

9. Prove that S^2 and $S^3 \times \mathbb{C}P^\infty$ have isomorphic homotopy groups but are not weakly homotopy equivalent.

Answer. From the Hopf bundle, we have $\pi_n(S^2) = \pi_n(S^3)$ for all n except $n = 2$, for which $\pi_2(S^3) = 0$ while $\pi_2(S^2) = \mathbb{Z}$. On the other hand, $\mathbb{C}P^\infty$ is $K(\mathbb{Z}, 2)$ which implies that $S^3 \times \mathbb{C}P^\infty$ has the same homotopy groups as S^2 .

Yet, the 3-spheroid $S^3 \rightarrow S^2 \rightarrow S^3 \times \mathbb{C}P^\infty$, where the first map is the Hopf bundle and the second is any map f , is contractible. Indeed, $S^3 \rightarrow \mathbb{C}P^\infty$ is contractible because $\pi_3(\mathbb{C}P^\infty) = 0$, and $S^3 \rightarrow S^2 \rightarrow S^3$ is contractible because $S^2 \rightarrow S^3$ is (by the cellular approximation theorem). Thus, $f_* : \pi_3(S^2) \rightarrow \pi_3(S^3 \times \mathbb{C}P^\infty)$ maps the generator to 0, showing that no f is a homotopy equivalence, and by Whitehead's theorem, implies that the two spaces are not weakly homotopy equivalent.

10. Let X be a $K(G, n)$, and Y be a cellular $K(H, n)$. Show that a map $Y \rightarrow X$ inducing a given group homomorphism $\phi : H \rightarrow G$ exists, and is unique up to homotopy.

Answer. Since $\pi(Y, X) = \pi(Y', X)$ when Y' is cellular and homotopy equivalent to Y , we may assume that Y is obtained from the bouquet $sk_n Y := \bigvee_\alpha S_\alpha^n$ of spheres labeled by generators of H by attaching cells of higher dimensions (of dimension $n + 1$ according to a set of relations of H , and then of dimension $> n + 1$ killing all higher homotopy groups). Then we can define $f_n : sk_n Y \rightarrow X$ by mapping each sphere S_α^n according to a spheroid representing $\phi(\alpha) \in \pi_n(X, x_0)$. This map extends to $f_{n+1} : sk_{n+1} Y \rightarrow X$ because the attaching map $\partial D^{n+1} \rightarrow sk_n Y$ for each $n + 1$ -cell represents the trivial element in $H = \pi_n(Y)$, hence its image under f_n is also trivial in $G = \pi_n(X)$, meaning that it is extendable to D^{n+1} . The inductive extension

of f_{n+1} to maps $f_{n+k} : sk_{n+k}Y \rightarrow X$ with $k > 1$ does not meet any obstruction because any spheroid of dimension $> n$ in X is contractible.

To show that two maps $f, g : Y \rightarrow X$ inducing the same homomorphism $\phi : \pi_n(Y) \rightarrow \pi_n(X)$ are homotopic, we apply cell induction. At the base, we may assume that the homotopy (i.e. a path in X) between the base points $f(y_0)$ and $g(y_0)$ has been selected. At the step of induction we want to extend an already constructed homotopy $h : sk_{k-1}Y \times [0, 1] \rightarrow X$ between $f|_{sk_{k-1}Y}$ and $g|_{sk_{k-1}Y}$ to an k -cell D^k of Y with the characteristic map $\Phi : D^k \rightarrow sk_kY$. This amounts to extending to $D^k \times [0, 1]$ a map to X equal to $f \circ \Phi$ on $D^k \times 0$, $g \circ \Phi$ on $D^k \times 1$, and $h \circ \Phi|_{\partial D^k}$ on $\partial D^k \times [0, 1]$. All these maps combined define a k -spheroid $\partial(D^k \times [0, 1]) \rightarrow X$, and its extension to $D^k \times [0, 1]$ exists iff the spheroid is contractible in X . This is the case for all $k \neq n$ because $\pi_k(X) = 0$. For $k = n$ this is also true because the restrictions of f and g to the n -cells $(D^n/\partial D^n)$ of sk_nY represent the same element of $G = \pi_n(X)$, and are therefore homotopic. [There is a little subtlety in the case $n = 1$ when the identification of $\pi_1(X)$ with G depends on a path connecting the base points. So, we assume that “represent the same element” refers to the identification using the path between $f(y_0)$ with $g(y_0)$ selected at the base of induction.)

11. Show that the tautological embedding of $\mathbb{C}P^\infty$ into $G_+(\infty, 2)$ is a homotopy equivalence.

Answer. A complex line $L \subset \mathbb{C}^\infty \cong \mathbb{R}^{2\infty}$ is a real plane equipped with the complex orientation (which is defined by considering the multiplication by $\sqrt{-1}$ in $L \cong \mathbb{R}^2$ as the counter-clockwise rotation in). This defines an embedding of $\mathbb{C}P^\infty$ into $G_+(2\infty, 2)$.

The spaces are respectively the classifying spaces BU_1 and BSO_2 for actually the same group: $U_1 \cong SO_2$. Moreover our embedding induces the universal U_1 -bundle from the universal SO_2 -bundle. Namely the right-oriented orthonormal bases in L considered as a Euclidean plane have the form $(u, \sqrt{-1}u)$ where u is a unit vector in L (i.e. they are one-to-one correspondence with unitary bases in L). This implies that the embedding is a weak homotopy equivalence, and hence a homotopy equivalence since both spaces are CW-complexes.

12. Prove that a continuous group homomorphism $f : G \rightarrow G'$ induces a map $BG \rightarrow BG'$, which is a weak homotopy equivalence provided that f is.

Answer. In terms of transition functions $\phi_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow G$, any G -bundle defines a G' -bundle using transition functions $f \circ \phi_{\alpha\beta}$. This functor

must be represented by a map $BG \rightarrow BG'$, namely by the map inducing from the universal G' -bundle the G' -bundle over (cellular) BG associated this way with the universal G -bundle.

Moreover, the maps $(\text{id}, f) : U_{\alpha/\beta} \times G \rightarrow U_{\alpha/\beta} \times G'$ between local trivializations over U_α, U_β , over their intersection commute with the gluing maps $(x, g) \mapsto (x, \phi_{\alpha\beta}(x)g)$ and $(x, g') \mapsto (x, f(\phi_{\alpha\beta}(x))g')$. This shows that f induces a fiberwise map between the G - and G' - fibrations over BG .

Alternatively, the same conclusion can be obtained by upgrading $f : G \rightarrow G'$ to the map $F : EG \times_G G \rightarrow EG \times_G G'$. These are the total spaces of the bundles over BG obtained from the universal principal bundle $EG \xrightarrow{G} BG$ by replacing the fiber G with “ G -spaces”, G and G' respectively, where $G \ni g$ acts on G and G' via left translations by respectively g and $f(g)$. Since left and right translations commute, the groups G and G' still act freely via right translations respectively on $EG \times_G G$ (which is of course not different from EG) and on $EG \times_G G'$. Consequently we have over $BG = EG/G$ the principal G - and G' -bundles with an equivariant map F between them which is fiberwise equivalent to f : $F(yg) = F(y)f(g)$.

Another construction can be based on Milnor’s models: the join of infinitely many copies of G is mapped to the join of infinitely many copies of G' : $F : \{(t_k, g_k)\} \mapsto \{(t_k, f(g_k))\}$, and this map is “ f -equivariant”: $F(yg) = F(y)f(g)$, thereby inducing a map between the universal bundles over BG and BG' fiberwise identical to $f : G \rightarrow G'$ (up to left translations).

In either case, when f is a WHE, it follows from the 5-lemma applied to the morphism induced by F between the exact homotopy sequences of the bundles that (in the first construction) the space $EG \times_G G'$ has trivial homotopy groups making the G' -bundle over BG universal (and making therefore $BG' = BG$), or (in Milnor’s construction) that $BG \rightarrow BG'$ is a WHE.

13. Classify principle $SL_2(\mathbb{C})$ -bundles over $\mathbb{C}P^2$.

Answer. $SL_2(\mathbb{C})$ retracts deformationally to $SU_2 = Sp_1$, whose classifying space is $\mathbb{H}P^\infty$. By cell approximation, $\pi(\mathbb{C}P^2, \mathbb{H}P^\infty) = \pi(\mathbb{C}P^2, S^4)$ since $S^4 = \mathbb{H}P^1$ is the 5-skeleton (or even 7-skeleton) of $\mathbb{H}P^\infty$ (consisting of one cell in each dimensions 0,4,8,12,...). By Borsuk’s theorem, one may assume without loss of generality that a map $\mathbb{C}P^2 \rightarrow S^4$ is homotopic factors through the projection $p : \mathbb{C}P^2 \rightarrow \mathbb{C}P^2/\mathbb{C}P^1 = S^4$, and is therefore homotopic to the composition of p with a map $S^4 \rightarrow S^4$ of degree $d \in \mathbb{Z} = \pi_4(S^4)$. To show that d is invariant under homotopies of maps $\mathbb{C}P^2 \rightarrow S^4$ (and not merely $\mathbb{C}P^2/\mathbb{C}P^1 \rightarrow S^4$), we note that (whereas

$\pi_4(\mathbb{C}P^2) = \pi(\mathbb{C}P^\infty) = 0$ the above composition induces multiplication by d in homology: $H_4(\mathbb{C}P^2) = \mathbb{Z} \xrightarrow{d} \mathbb{Z} = H^4(S^4)$.

14. Let X be the mapping torus of a degree-2 map of (S^2, pt) to itself (i.e. obtained from the cylinder $S^2 \times [0, 1]$ by gluing its boundaries by this degree-2 map). Show that the embedding of $i : S^1 \subset X$ as the interval $pt \times [0, 1]$ glued at the endpoints induces isomorphism in homology but is not a homotopy equivalence.

Answer. The cellular chain complex of X is generated by cells e^0, e^1, e^2, e^3 of dimensions 0, 1, 2, 3 respectively which are mapped to pt , $pt \times S^1$, $S^2 \times 0$ and the bulk of X . The boundary $\partial e^3 = 2e^2 - e^2 = e^2$, while all other cells have zero boundaries in the complex. Thus, the homology of X comes from the trivial subcomplex spanned by e^0 and e^1 , which shows that $i_* : H_*(S^1) \rightarrow H_*(X)$ is an isomorphism.

Obviously, the inclusion also induces the isomorphism of the fundamental groups. However, the universal cover of S^1 is contractible, while the universal cover of X can be described as a sequence of cylinders $S^2 \times [n-1, n]$ whose boundary $S^2 \times n$ is attached to the boundary $S^2 \times n$ of the next cylinder $S^2 \times [n, n+1]$ by the degree-2 map. Therefore $\pi_2(X)$ can be identified with the group $\mathbb{Z}[1/2]$ of binary rationals $m/2^k$ (the same as the fundamental group of studied in Problem 5).

15. Prove that each skeleton $sk_n X$ of a contractible CW-complex X is homotopy equivalent to a bouquet of n -spheres.

Solution. For $n = 0$, the skeleton is a discrete set and hence a bouquet of 0-spheres. For $n = 1$, it is a graph, connected since $\pi_0(sk_1 X) = \pi_0(X)$, and hence homotopy equivalent to a bouquet of circles as in the argument above. Since in general $\pi_k(X)$ is determined by the $(k+1)$ -skeleton of X , we conclude that $sk_n X$ is $(n-1)$ -connected. For $n > 1$, the Hurewicz theorem implies that $H_k(sk_n X) = 0$ for all $k < n$, and $\pi_n(sk_n X) = H_n(sk_n X)$. The latter is a free abelian group, since for an n -dimensional cell space, it is the kernel of the boundary operator in the cellular chain complex, and hence a subgroup in the free abelian group of chains. Now take a bouquet Y_n on n -spheres labeled by a set of free generators of $\pi_n(sk_n X)$, and map it to $sk_n X$ by the bouquet of spheroid maps realizing the generators. This map induces isomorphism in homology, and hence is a homotopy equivalence by Whitehead's homological theorem.

16. Compute $H^*(\tilde{X})$ where $\tilde{X}$ is the universal cover of the mapping torus space X from problem 14.

Answer. The universal cover is simply connected and hence $H_0(\tilde{x}) = \mathbb{Z}$, $H_1(\tilde{X}) = 0$, and by the universal coefficient formulas $H_0(\tilde{x}) = \mathbb{Z}$, $H_1(\tilde{X}) = 0$ as well. As we know from the solution of problem 14, $\pi_2(\tilde{X}) = \mathbb{Z}[1/2]$, the additive group of binary rationals, and so is $H_2(\tilde{X})$ by the Hurewicz theorem. The group $\mathbb{Z}[1/2]$ is the direct limit of the nested sequence $\mathbb{Z} \subset \frac{1}{2}\mathbb{Z} \subset \frac{1}{4}\mathbb{Z} \subset \dots$. To apply the universal coefficient theorem, we can take its free resolution $0 \rightarrow F_1 \rightarrow F_0 \rightarrow 0$ where $F_1 = F_0 = \oplus_{n \geq 0} \mathbb{Z}$, the generators f_n of F_0 are mapped to $1/2^n$, and the generators e_n of F_1 are mapped to $f_n - 2f_{n+1}$. Applying $\text{Hom}(-, \mathbb{Z})$ we obtain an exact sequence

$$0 \leftarrow \text{Ext}(\mathbb{Z}[1/2], \mathbb{Z}) \leftarrow \text{Hom}(F_1, \mathbb{Z}) \leftarrow \text{Hom}(F_0, \mathbb{Z}) \leftarrow \text{Hom}(\mathbb{Z}[1/2], \mathbb{Z}) \leftarrow 0,$$

where $\text{Hom}(\mathbb{Z}[1/2]) = 0$. A homomorphism of $\phi : \oplus_{n \geq 0} \mathbb{Z} \rightarrow \mathbb{Z}$ is determined by the sequence $(\phi(f_0), \phi(f_1), \dots)$ of integers. The middle arrow $\text{Hom}(F_0, \mathbb{Z}) \rightarrow \text{Hom}(F_1, \mathbb{Z})$ can be described as the map of sequences $(a_0, a_1, \dots) \mapsto (b_0, b_1, \dots) = (a_0 - 2a_1, a_1 - 2a_2, \dots)$. To understand the cokernel of this homomorphism, denote by $\mathbb{Z}_{(2)}$ the group (actually) ring of 2-adic integers. By definition it consists of usual binary integers with the operations performed according to the standard algorithms of binary addition and multiplication (with the usual binary carry-overs), except that the binary sequences don't have to be finite, but can spread infinitely to the left. Since the standard algorithms are performed "from right to left", they still work. Considering first the b -sequences modulo a -sequences with $a_0 = 0$, we can identify the quotient with $\mathbb{Z}_{(2)}$ by the map

$$(b_0, b_1, \dots) \mapsto b_0 + 2b_1 + 4b_2 + \dots + 2^n b_n + \dots.$$

Indeed, the series converges in $\mathbb{Z}_{(2)}$, and the sum is 0 exactly when for $b_0 \equiv 0 \pmod{2}$, $b_0 + 2b_1 \equiv 0 \pmod{4}$, etc., i.e. when for some $(a_1, a_2, \dots)$ we have $b_0 = -2a_1$, and $b_0 + 2b_1 = -4a_2$, and so on. Finally taking the quotient by the image $\mathbb{Z} \subset \mathbb{Z}_{(2)}$ of the sequences $(a_0, 0, 0, \dots)$, we conclude that $\text{Ext}(\mathbb{Z}[1/2], \mathbb{Z}) = \mathbb{Z}_{(2)}/\mathbb{Z}$, the additive group of 2-adic integers modulo the ordinary ones. Thus, $H^2(\tilde{X}) = 0$, and $H^3(\tilde{X}) = \mathbb{Z}_{(2)}/\mathbb{Z}$.

Remark. Refraining from the use of the universal coefficient theorem and instead dualizing the cellular homological complex of $\tilde{X}$ results in essentially the same computation because the complex provides a free resolution for $\mathbb{Z}[1/2]$.

17. Let $M^n \subset S^N$ be a closed oriented submanifold in an (oriented) sphere, and let U be a tubular neighborhood of M in S^N . Show that the composition

$$H_N(S^N) \rightarrow H_N(S^N, S^N - U) = H_N(U, \partial U) \xrightarrow{\cap \mathbf{t}} H_n(U) = H_n(M)$$

of the natural projection with the Thom isomorphism maps the generator $[S^N] \in H_N(S^N)$ to a fundamental class $[M] \in H_n(M)$ (i.e. that for every $x \in M$ the projection $H_n(M) \rightarrow H_n(M, M - x) = \mathbb{Z}$ maps this class to 1).

Answer. Let $D_x^{N-n} \times D_x^n \rightarrow D_x^n$ be a trivialization of the tubular neighborhood U (considered as a bundle over M) over a small disk D_x^n in M centered at x . The class $[S^N]$ projects to the generator in

$$H_N(S^N, S^N - (D_x^{N-n} \times D_x^n)) = H_N(D_x^{N-n} \times D_x^n, \partial(D_x^{N-n} \times D_x^n)) = \mathbb{Z}.$$

The Thom class $\mathbf{t} \in H^{N-n}(U, \partial U) = H_N(S^N, S^N - U)$ evaluates to 1 on the (oriented) fibers of the tubular neighborhood (considered as relative cycles in $H_{N-n}(U, \partial U)$), and in particular on the cycle $[D_x^{N-n}]$ representing the generator in $H_{N-n}(D_x^{N-n}, \partial D_x^{N-n}) = \mathbb{Z}$. Due to the naturality of the cap-product (and excision), the image of $[M] := [S^N] \cap \mathbf{t}$ in $H_n(D_x^n, \partial D_x^n) = \mathbb{Z}$ coincides with the cap-product of the generator in $H_N(D_x^{N-n} \times D_x^n, \partial(D_x^{N-n} \times D_x^n))$ with the generator in $H^{N-n}(D_x^{N-n}, \partial D_x^{N-n})$. Due to the “key example” from the lectures (or, equivalently, from Theorem on p. 206 from Fomenko-Fuchs about the product of two simplexes), this cap product is equal to the generator in $H_n(D_x^n, \partial D_x^n)$ as required.

18. A fiberwise map $f : E \rightarrow E'$ between (homologically simple) fibrations $p : E \rightarrow B$ and $p' : E' \rightarrow B'$ inducing a *cellular* map $g : B \rightarrow B'$ between CW-bases respects the filtrations of the total spaces by E_p and E'_p , and hence defines a morphism between the spectral sequences of the fibrations, with the homomorphisms $(f_*)_{pq}^2 : H_p(B; H_q(F)) \rightarrow H_p(B'; H_q(F'))$ induced by g and the restriction $F \rightarrow F'$ of f to the fibers. Prove that even when the map $g : B \rightarrow B'$ is not cellular, the morphism between the spectral sequences is well-defined starting the term E^2 .

Suggestion: Consider the fibration $\mathcal{E} \rightarrow B \times [0, 1]$ with the fiber F' induced by the homotopy $G : B \times [0, 1] \rightarrow B'$ between $g_0 = g$ and its cellular approximation g_1 , and examine the morphisms between spectral sequences induced by fibered maps: $E \rightarrow \mathcal{E}_0$ (over $B \times 0$), $\mathcal{E}_1 \rightarrow \mathcal{E}$, and $\mathcal{E}_1 \rightarrow E'$ (all fibered over cellular maps id_B , $B \times 1 \subset B \times [0, 1]$ and $g_1 : B \times 1 \rightarrow B'$ of the respective bases).

Remark. For Hurewicz fibrations, the construction is simpler due to the Covering Homotopy Property valid for maps of arbitrary (rather than cellular) spaces.

Answer. The map $E \rightarrow \mathcal{E}_0$ fibered over the identity map $B \rightarrow B \times 0$ and induced by the initial map $f : E \rightarrow E'$ defines a morphism between the spectral sequences with the homomorphism between the E^2 terms $H_p(B; H_q(F)) \rightarrow H_p(B; H_q(F'))$ induced by the restriction $F \rightarrow F'$ of f to the fibers.

For the restriction of the induced F' -fibration $\mathcal{E} \rightarrow B \times [0, 1]$ to $B \times 1$ (and to $B \times 0$) the morphism between the spectral sequences is the isomorphism at the term $E_{pq}^2 = H_p(B; H_q(F'))$ induced by the natural maps of the bases and the fibers. The isomorphism commutes with the differentials d_{pq}^2 (hence identifying them), and therefore the terms E^3 are also the same, and for the same reason the isomorphism persists in all subsequent terms E^r .

The fibered map $\mathcal{E}_1 \rightarrow B$ from $E' \rightarrow B'$ induced by the cellular $g_1 : B \rightarrow B'$ defines a morphism between the spectral sequences with the homomorphism between the E^2 -terms $g_{1*} : H_p(B; H_q(F')) \rightarrow H_p(B'; H_q(F'))$.

The composition of these three morphisms (with the middle one being an isomorphism) defines the required morphism between the spectral sequences (starting the term E^2) which does not depend on the cellular structures of the bases (showing in particular that the spectral sequence of a Serre fibration does not depend on the CW-structure of the base starting the term E^2).

Moreover, since each of the three morphisms is induced by a map of filtered spaces, the homomorphism of the E^∞ -terms is obtained by applying the “Gr” operation to the homomorphism $f_* : H_n(E) \rightarrow H_n(E')$ of filtered homology groups (although the filtrations were initiated by different cell structures on the bases). Consequently, the filtration in the homology of the total space of a fibration does not depend on the cell structure of the base.

19. In $\mathbb{C}P^2 \times \mathbb{C}P^2$ consider the hypersurface F given by $x_1y_1 + x_2y_2 + x_3y_3 = 0$ where (x_1, x_2, x_3) and (y_1, y_2, y_3) are homogeneous coordinates on the left and right projective planes respectively. Identify F with the manifold of complete flags in $\mathbb{C}^3$, find the kernel of the homomorphism $\mathbb{Z}[u, v]/(u^3, v^3) \rightarrow H^*(F)$, where u and v are the generators in the cohomology algebras of left and right projective planes Poincaré-dual to projective lines therein, and show that the homomorphism is surjective.

Hint: A possible approach is to use intersections of submanifolds in lieu of cohomological cup-product.

Answer. To a flag $L(\text{ine}) \subset P(\text{lane}) \subset \mathbb{C}^3$ one can associate $L \in \mathbb{C}P^2$ and $P \in \mathbb{C}P^{2*}$ (projectivization of the dual space to $\mathbb{C}^3$). If a line L is spanned by vector (x_1, x_2, x_3) and a plane P is given by linear equation with coefficients (y_1, y_2, y_3) , then the condition $L \subset P$ is given by the equation $x_1y_1 + x_2y_2 + x_3y_3 = 0$.

Recall that the Poincaré polynomial of the flag manifold is $1 + 2q + 2q^2 + q^3$.

With this understanding, the cohomology classes of $F = \{(L, P) | L \subset P\}$: are Poincaré-dual to the following complex submanifolds:

- u : L lies in a fixed plane P_0 ;
- v : P contains a fixed line L_0 ;
- u^2 : $L = L_0$;
- v^2 : $P = P_0$;
- uv : L lies in P_0 and P contains L_0 (not contained in P_0 , i.e. $P = L \oplus L_0$);
- $v^2u = pt$: P is a fixed plane and L is the intersection with another fixed plane;
- $u^2v = pt$ (likewise);
- $u^3 = v^3 = 0$ (because this is so in $\mathbb{C}P^2 \times \mathbb{C}P^2$).

Thus, we have the following intersection indexes: $(u, v^2) = (v, u^2) = (u, uv) = (v, uv) = 1$ and $(u, u^2) = (v, v^2) = 0$. Consequently: u and v are linearly independent and form a $\mathbb{Z}$ -basis in $H^2(F)$; u^2 and v^2 are linearly independent and form a $\mathbb{Z}$ -basis in $H^4(F)$; $u^2 + v^2 = uv$ (as they have the same intersection indexes with the basis in $H^2(F)$; $u^2v = uv^2$). Thus, $H^*(F) = \mathbb{Z}[u, v]/(u^2 + v^2 - uv, u^2v - uv^2)$ (of which $u^3 = v^3 = 0$ are actually consequences).

Remark. It follows from the fact that the line bundles L , P/L , and $\mathbb{C}^3/P$ add up to the trivial bundle $\mathbb{C}^3$ that their first Chern classes a, b, c satisfy: $(1 + a)(1 + b)(1 + c) = 0$, i.e. $a + b + c = 0$, $ab + bc + ca = 0$, and $abc = 0$. In fact $a = -u$, $c = v$, and $b = -a - c = u - v$, and so we obtain the same relations as above. More generally, it follows from the

spectral sequence of the fibration $BT^n \rightarrow BU_n$ that the cohomology of the fiber $H^*(U_n/T^n) = \mathbb{Z}[x_1, \dots, x_n]/(\sigma_1(x), \dots, \sigma_n(x))$.

20. Show that the tangent bundle T to $\mathbb{R}P^n$ can be described as $Hom(L, \mathbb{R}^{n+1}/L)$ where L is the tautological (Möbius) line bundle, use this to compute Stiefel-Whitney classes of T , and conclude that they all are zeroes when $n+1$ is a power of 2.

Answer. On a grassmannian $G(m, k)$, chart near a given point $\mathbb{R}^k \subset \mathbb{R}^m$ can be described by interpreting linear subspaces as graphs of linear maps from $\mathbb{R}^k$ to its orthogonal complement. In particular, the tangent space at the center of this chart is identified with $Hom(\mathbb{R}^k, \mathbb{R}^m/\mathbb{R}^k)$. Thus, the tangent bundle T to $\mathbb{R}P^n$ is identified with $Hom(L, \mathbb{R}^{n+1}/L)$ indeed. In other words, we have an exact sequence of bundles:

$$0 \rightarrow L^* \otimes L \rightarrow L^* \otimes \mathbb{R}^{n+1} \rightarrow T \rightarrow 0.$$

Since $L^* \otimes L = Hom(L, L) = \mathbb{R}$ is trivial, we conclude that the total Stiefel-Whitney class $w(T) = w(L^*)^{n+1} = (1+t)^{n+1} \in \mathbb{Z}_2[t]/(t^{n+1})$, where t is the generator in $H^2(\mathbb{R}P^n; \mathbb{Z}_2) = \mathbb{Z}_2$. In particular, $(1+t)^{2^r} \equiv 1+t^{2^r} \pmod{2} \equiv 1 \pmod{t^{2^r}}$.

21. Show that composition of loops $\Omega K(G, n+1) \times \Omega K(G, n+1) \rightarrow \Omega K(G, n+1)$ induces the usual addition operation in cohomology $H^n(X; G) = \pi(X, K(G, n)) = \pi(X; \Omega K(G, n+1))$ of a CW-complex X .

Answer. Let us use the short-cut notation $K := K(G, n) \sim \Omega K'$ where $K' := K(G, n+1)$. Note that (for any space K') the composition of loops $\Omega K' \times \Omega K' \rightarrow \Omega K'$ composed with either of the two embeddings of $\Omega K'$ into the product (as $\Omega K' \times pt$ or $pt \times \Omega K'$ where pt is the constant loop at the base point of K') is homotopic to the identity on $\Omega K'$. (Namely, the composition of a loop γ with the constant loop pt is reparameterized into γ in a standard way applicable to all loops γ simultaneously.) Therefore the map $c : K \times K \rightarrow K$ corresponding to the composition of loops on $\Omega K' \sim K$ composed with either of the two coordinate embeddings of $i_{1,2} : K \rightarrow K \times K$ is also homotopic to the identity on K . Assuming K cellular, recall that the fundamental class $F \in H^n(K; G) = Hom(G, G)$ is represented by the identity map from K to itself. The previous observations shows that $i_1^* c^*(F) = F$ and $i_2^* c^*(F) = F$. We have: $H^n(K \times K; G) = Hom(G \otimes \mathbb{Z} \oplus \mathbb{Z} \otimes G, G)$ due to the universal coefficient formula for G -coefficients and the homological Künneth formula for $\mathbb{Z}$ -coefficients. Therefore $c^*(F) = F \oplus F$. This is all we need, since for any

cellular X and any maps $f_1, f_2 : X \rightarrow K$ (i.e. for $f_1 \times f_2 : X \rightarrow K \text{ times } K$), we will have

$$(f_1 \times f_2)^* c^*(F) = (f_1 \times f_2)^*(F \oplus F) = f_1^*(F) + f_2^*(F).$$

22. Show that the 1st Chern class of a complex n -dimensional vector bundle V over a CW-complex B (defined by means of the obstruction theory) coincides with the 1st Chern class of its “determinant” line bundle $\wedge^n V$.

Answer. The 1st Chern class is defined as the 1st obstruction to a unitary n -frame in V over the 2-skeleton B^2 . From the Schubert cell decomposition of the classifying space $\mathbb{C}G(\infty, n)$, we know that its 3-skeleton is $\mathbb{C}P^1 \subset \mathbb{C}G(\infty, 1)$. Applying the cellular approximation theorem to the map $B^2 \rightarrow \mathbb{C}G(\infty, n)$ inducing the restriction of V to B^2 , we conclude that this restriction is equivalent to the sum $L \oplus \mathbb{C}^{n-1}$ of a complex line bundle with the trivial $n - 1$ -dimensional one. The “determinant” line bundle $\wedge^n(L \oplus \mathbb{C}^{n-1})$ coincides with L . On the other hand, the problem of constructing a unitary n -frame in $L \oplus \mathbb{C}^{n-1}$ reduces to finding a unit section of L , i.e. both $c_1(V)$ and $c_1(\wedge^n V)$ are the 1st obstructions to finding a unit section of L over B^2 .

23. Let $t \in \tilde{H}^k(\hat{E})$ be the Thom class of an oriented k -dimensional vector bundle $E \rightarrow B$ over a CW-complex B . Prove that the restriction of t to the zero section is the Euler class of the bundle. Derive from this that if $E \rightarrow B$ is smooth fibration over a closed (oriented) manifold, then the fundamental class of zero locus $s^{-1}(0)$ of a smooth section $s : B \rightarrow E$ transverse to the zero section is Poincaré-dual in B to the Euler class of the bundle.

Answer. Let s be a non-vanishing section of the bundle over the $k - 1$ -dimensional skeleton B^{k-1} extended to B^k to a section transverse to the zero section inside every k -dimensional cell. The algebraic number of zeroes of s inside a given cell (i.e. $\sum \pm 1$ over such zeroes with the signs defined by orientations) is the value of the Euler obstruction cochain on this cell. On the other hand, the section s is homotopic to the zero section, and we can use it instead of the zero section to compute the restriction of the Thom class to B^k . Note however that $t \in H^k(E, E - B)$, and since s lands in $E - B$ on B^{k-1} , the restriction of t to $s(B^k)$ lies in $H^k(B^k, B^{k-1})$. It is clear now that the Thom class, characterized by taking the value $+1$ on each oriented fiber of the vector bundle, induces in $H^k(B^k, B^{k-1})$ the Euler cochain. Indeed, modulo $E - B$, the section s on a given k -cell is homotopic to the algebraic $\sum \pm [\mathbb{R}^k]$ of the fibers over the zeroes of s inside that cell, and the signs are prescribed by the orientations as before.

Finally, in the manifold case, we use the homological (intersection theory) description of cohomological pull-back. The Poincaré-dual of $t \in H^k(E, E - B)$ is the fundamental class $[B]$ of the zero section. Respectively the pull-back $i^*t = e$ of the Thom class under the embedding $i : B \subset E$ of the zero section is Poincaré-dual to the fundamental class of the transversal self-intersection of the zero section (i.e. of the intersection of the zero section with another smooth section transverse to it).

24. Use multiplication in the spectral sequence of the fibration $pt \xrightarrow{K(\mathbb{Z}, 2)} K(\mathbb{Z}, 3)$ to compute the group $H^6(K(\mathbb{Z}, 3))$ and identify in it the square e^2 of the fundamental class $e \in H^3(K(\mathbb{Z}, 3))$.

Answer. Since $K(\mathbb{Z}, 2) = \mathbb{C}P^\infty$, in the fiber of the cohomological spectral sequence of the fibration $pt \xrightarrow{K(\mathbb{Z}, 2)} K(\mathbb{Z}, 3)$ we find $\mathbb{Z}[x]$, where x is the fundamental class of $K(\mathbb{Z}, 2)$. The next two columns in the $E_2 = E_3$ -term are zeros, and the column with $p = 3$ contains the fundamental class e ($q = 0$), 0 ($q = 1$), and xe ($q = 2$), which generate the respective groups $E_2^{3,q}$. We have $d_3(x) = e$, since it serves as the last chance for both x and e to disappear from $E_\infty = H^*(pt)$. In the complex

$$0 \rightarrow E_3^{0,4} \xrightarrow{d_3} E_3^{2,3} \xrightarrow{d_3} E_3^{6,0} \rightarrow 0,$$

whose first two terms are spanned by x^2 and xe respectively, we have $d_3(x^2) = 2xe$, and $d_3(xe) = e^2$. If it were not exact, the remaining cohomology would survive till E_∞ yielding a contradiction. Thus, e^2 is the only non-zero element in $H^6(K(\mathbb{Z}, 3); \mathbb{Z}) = \mathbb{Z}_2$. Besides, for $p = 4, 5$, the groups $H^p(K(\mathbb{Z}, 3)) = 0$ since x^2 (which could potentially kill something in $H^5(K(\mathbb{Z}, 5))$) doesn't survive past E_3 .

25. Show that $H^*(\mathbb{C}G(n, k)) = \mathbb{Z}[c'_1, \dots, c'_k, c''_1, \dots, c''_{n-k}]/I$, where c'_i and c''_j are the Chern classes of the tautological $\mathbb{C}^k$ - and $\mathbb{C}^{n-k} := \mathbb{C}^n/\mathbb{C}^k$ -bundles, and the ideal I is generated by the homogeneous components of the relation $(1 + c'_1 + \dots + c'_k)(1 + c''_1 + \dots + c''_{n-k}) = 1$. *Suggestion:* First compute the U_n -equivariant cohomology as an $H^*(BU_n)$ -algebra.

Answer. Since $\mathbb{C}G(n, k) = U_n/(U_k \times U_{n-k})$, we have

$$H_{U_n}^*(\mathbb{C}G(n, k)) = H^*(BU_k \times BU_{n-k}) = \mathbb{Z}[c'_1, \dots, c'_k] \otimes \mathbb{Z}[c''_1, \dots, c''_{n-k}],$$

with the structure of the module over $H^*(BU_n) = \mathbb{Z}[c_1, \dots, c_n]$ determined from the map $BU_k \times BU_{n-k} \xrightarrow{\mathbb{C}G(n, k)} BU_n$ (which induces the direct sum of

the tautological bundles from the one over BU_n):

$$(1 + c'_1 + \cdots + c'_k)(1 + c''_1 + \cdots + c''_{n-k}) = (1 + c_1 + \cdots + c_n).$$

From the Serre spectral sequence of the above fibration (where all degrees are even) we find $E_2 = E_\infty$ which is therefore a free $H^*(BU_n)$ -module, and hence so is $H_{U_n}^*(\mathbb{C}G(n, k))$. It follows that (as a module)

$$H^*(\mathbb{C}G(n, k)) = H_{U_n}^*(\mathbb{C}G(n, k))/(c_1, \dots, c_n)H_{U_n}^*(\mathbb{C}G(n, k)),$$

i.e. (as a ring) $H^*(\mathbb{C}G(n, k)) = \mathbb{Z}[c', c'']/(c_1, \dots, c_n)$.

26. Let x_1, x_2, x_3 be the T^3 -equivariant 1st Chern classes of line bundles $V^1, V^2/V^1, \mathbb{C}^3/V^2$ over the manifold F_3 of complete flags in $\mathbb{C}^3$. Write out the fixed point localization formula for the "integral" $\int_{F_3} P$ of a polynomial $P(x_1, x_2, x_3)$, and apply it in order to compute *Chern numbers* of F_3 , i.e. the values of the Chern classes c_1^3, c_1c_2, c_3 of the tangent bundle on the fundamental class. *Suggestion:* To find the T -equivariant Euler classes of the tangent spaces at each of the 6 fixed points, think of F_3 as the homogeneous space of $GL_3(\mathbb{C})/(\text{upper-triangular matrices})$.

Answer. Actually for any n , the T^n -localization formula of F_n is

$$\int_{U_n/T^n} P(x_1, \dots, x_n) = \sum_{\sigma \in S_n} \text{sign } \sigma \frac{P(-\lambda_{\sigma(1)}, \dots, -\lambda_{\sigma(n)})}{\prod_{i < j} (\lambda_i - \lambda_j)},$$

where $x_k := c_1^T(V^k/V^{k-1})$ are the T -equivariant 1st Chern classes and $\lambda_i := c_1(L_i^{-1}) \in H^*((\mathbb{C}P^\infty)^n)$, L_i being the Hopf bundle over the i -th factor. Namely, the fixed points are the $n!$ coordinate flags with $V^k := \text{Span}(e_{\sigma(1)}, \dots, e_{\sigma(k)})$. When $\sigma = \text{id}$, The fiber of V^k/V^{k-1} at this fixed point is spanned by e_k , the eigenvector of the torus of unitary diagonal matrices corresponding to the eigenvalue $e^{2\pi\sqrt{-1}\lambda_k}$. This explains why at this point $x_k = -\lambda_k$. The stabilizer $B \subset GL_n(\mathbb{C})$ of this flag consists of invertible upper-triangular matrices, and the tangent space $\text{Lie}GL_n(\mathbb{C})/\text{Lie}B$ at this point is identified with the space of lower-triangular matrices. The individual matrix elements δ_{ij} are eigenvectors of the adjoint action of the torus with the eigenvalues $e^{2\pi\sqrt{-1}(\lambda_i - \lambda_j)}$. This explains why the denominator $\prod_{i > j} (-\lambda_i + \lambda_j)$ is the equivariant Euler class of the tangent space at this fixed point. Equivalently, this means that the equivariant total Chern class of the tangent bundle is equal to $\prod_{i > j} (1 + x_i - x_j)$.

In particular, for $n = 3$ the Chern classes of the tangent bundle are

$$\begin{aligned} c_1^T &= 2(x_3 - x_1), & c_3^T &= (x_2 - x_1)(x_3 - x_2)(x_3 - x_1), \\ c_2^T &= (x_2 - x_1)(x_3 - x_2) + (x_2 - x_1)(x_3 - x_1) + (x_3 - x_2)(x_3 - x_1) \end{aligned}$$

From this, we can obtain

$$\int_{F_3} c_3^T = 6 \text{ (obviously), } \int_{F_3} (c_1^T)^3 = 48, \text{ and } \int_{F_3} c_1^T c_2^T = 24.$$

In fact it is *a priori* clear that the results lie in $H^*(BT^n) = \mathbb{Z}[\lambda]$, (i.e. the denominators cancel out), and therefore don't depend on λ (since the degrees $= \deg(\text{integrand}) - \dim(\text{manifold}) = 0$ in all the three cases). One can use this to simply replace λ_i by numbers, e.g. $\lambda_i = i$ for $i = 1, 2, 3$. Thus, $\int_{F_3} (c_1^T)^3 =$

$$2^3 \frac{(1-3)^3 - (2-3)^3 + (2-1)^3 - (3-1)^3 + (3-2)^3 - (1-2)^3}{(1-2)(1-3)(2-3)} = 8 \frac{-12}{-2}.$$

Also, $c_1^T c_2^T = 2c_3^T + (c_1^T)^3/4$, and therefore $\int_{F_3} c_1^T c_2^T = 12 + 48/4$.